Effects of Lisinopril and Nifedipine on the Progression to Overt Albuminuria in IDDM Patients With Incipient Nephropathy and Normal Blood Pressure

GAETANO CREPALDI, MD
QUIRICO CARTA, MD
GIACOMO DEFERRARI, MD
RUGGERO MANGILI, MD
RENZO NAVALESI, MD
FAUSTO SANTEUSANIO, MD
ANTONIO SPALLUTO, MD
ALESSANDRO VANASIA, MD
G. MARCO VILLA, MD
ROMANO NOSADINI, MD
THE ITALIAN MICROALBUMINURIA STUDY GROUP IN IDDM

OBJECTIVE — Intervention trials on renal function in IDDM patients with microalbuminuria (MA) should adopt the rate of decline of glomerular filtration rate (GFR) as an outcome measure. However, normotensive IDDM patients with MA show no change in GFR over a follow-up period of 10 years. Thus, in the present study, we used the cumulative incidence of progression to albuminuria (albumin excretion rate [AER] >200 μg/min) from MA as the primary endpoint and the yearly increase in AER at a rate of 50% above baseline as the secondary endpoint of renal function.

RESEARCH DESIGN AND METHODS — Ninety-two normotensive IDDM patients underwent double-blind, double-dummy treatment with either lisinopril or slow-release nifedipine in comparison with placebo. Ten patients discontinued the study during the 3-year follow-up period.

RESULTS — During the 3-year follow-up period, 7 of 34 placebo-treated (20.6%), 2 of 32 lisinopril-treated (6.3%), and 2 of 26 nifedipine-treated (7.7%) patients progressed to clinical albuminuria (Fisher's exact test, $P < 0.03$). Time-to-event analysis indicated a reduction in the risk of progression to macroalbuminuria of 58.1% (95% CI 27.8–68.4%) in the 32 patients on lisinopril ($P < 0.02$) and of 62.5% (95% CI 32.5–73.4%) in the 26 patients on nifedipine ($P < 0.02$) after adjustment for mean blood pressure, glycated hemoglobin, and baseline AER in comparison with the 34 patients on placebo. Baseline AER was 71 μg/min (range: 20.7–187.3) in progressors and 73 μg/min (range: 20.2–174.1) in nonprogressors (NS). The percentage of patients who showed a >50% yearly increase of AER above baseline values was significantly lower in the lisinopril group (13 of 32, 40.6%, $P < 0.02$), but not in the nifedipine group (15 of 26, 57.7%), than in the placebo group (23 of 34, 67.6%). The lisinopril group had significantly lower blood pressure values during follow-up than either the nifedipine ($P < 0.05$) or the placebo ($P < 0.01$) group.

CONCLUSIONS — Our data show that both lisinopril and nifedipine are effective in delaying the occurrence of macroalbuminuria in normotensive IDDM patients with MA. As overt proteinuria strongly predicts end-stage renal failure, both treatments appear capable of preventing such a complication in normotensive IDDM patients with MA. However, lisinopril appears more powerful in slowing the course of nephropathy.

From the Istituto di Medicina Interna, University of Padova and Sassari and National Research Center for the Study of Aging, Policlinico Universitario, Padova, Italy.

Address correspondence and reprint requests to Romano Nosadini, MD, Istituto di Medicina Interna, University of Padova and Sassari and National Research Center for the Study of Aging, Policlinico Universitario, Via Giustiniani 2, 35128 Padova, Italy.

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Abbreviations: AER, albumin excretion rate; BP, blood pressure; dBP, diastolic blood pressure; GFR, glomerular filtration rate; MA, microalbuminuria; RR, relative risk; sBP, systolic blood pressure.

Captopril, an ACE inhibitor, has been shown to decrease albumin excretion rate (AER), to preserve glomerular filtration rate (GFR), and to lower the rate of development of overt nephropathy in hypertensive proteinuric IDDM patients (1). With regard to calcium channel-blockers, the Melbourne Diabetic Nephropathy Study has shown that nifedipine ameliorated renal function in a fashion similar to that of perindopril in IDDM patients with incipient nephropathy (2).

Microalbuminuria (MA) is observed in as many as 15–25% of IDDM patients in Europe, even in the absence of arterial hypertension (3,4). It is not clear whether MA predicts overt nephropathy to the same extent when associated with arterial hypertension. Normotensive microalbuminuric IDDM patients show no change in GFR over a follow-up period of 5–10 years, irrespective of antihypertensive treatment (5). Therefore, GFR does not appear to provide useful information on the effects of intervention trials in normotensive microalbuminuric IDDM patients.

Alternatively, two further parameters have been suggested as outcome measures of renal function in IDDM patients with MA and normal blood pressure (BP). First, the change of category from MA to macroalbuminuria is an important clinical event, as it indicates the occurrence of overt nephropathy, which irreversibly antedates end-stage renal failure (6,7). Second, an accelerated rate of increase of AER per year above baseline, although within the range of MA, has also been used to identify those microalbuminuric NIDDM patients likely to progress toward overt nephropathy (8).

In the present study, we used the rate of progression of AER from MA to macroalbuminuria to evaluate the course of renal function in normotensive IDDM patients during treatment with either ACE inhibitors or calcium channel blockers.

RESEARCH DESIGN AND METHODS

Eligibility (visit 1)

Fourteen hospital-based and diabetes outpatient centers in Italy participated for the current study. This was a double-blind, randomized, parallel-group, multicenter study. Patients were randomized to receive, after a 4-week placebo run-in, lisinopril, nifedipine, or placebo for 36 months.

For inclusion in the study, patients had to satisfy the following criteria in agreement with the National Diabetes Data Group (9): between 18 and 65 years of age; onset of IDDM before age 35, and insulin treatment within 3 years of diagnosis; clinical stability, as shown by glycated hemoglobin (HbA_{1c}) values <11% at entry and within 30% of the value at entry during the past 12 months; standing systolic BP (sBP) ≥115 and ≤140 mmHg (without antihypertensive treatment) and diastolic BP (dBP) ≥75 and ≤90 mmHg at entry; median AER value between 20 and 200 μg/min from three timed overnight urine collections at a routine control during the year before the trial (as measured in each individual peripheral participating center) and between 20 and 200 μg/min from three timed overnight urine collections within 2 weeks of entry (as measured in the central laboratory: R. Mangili, S. Raffaele, Milan, Italy) at visit 3, the inclusion visit; and GFR ≥80 ml · min^{-1} · 1.73 m^2 at visit 2 (randomization).

Any of the following was a criterion for exclusion from the trial: impaired renal function, defined as serum creatinine >10% above the upper limit of the normal laboratory range (125 μmol/l) and median AER rate >200 μg/min (according to the central laboratory measurement) at entry at visit 3, after randomization; history of any nondiabetic renal disease; hematuria; evidence of clinically significant liver or hematological disease; evidence of aortic or mitral valve obstruction, arrhythmias, unstable angina, or a history of myocardial infarction within the previous 3 months; clinical evidence of autonomic neuropathy; systemic malignancy; hyperkalemia (serum potassium >5.5 mmol/l at either pretrial screen or entry); serum triglycerides >3.4 mmol/l or total cholesterol >6.5 mmol/l at routine pretrial check; known familiar lipid disorders; known risk of transmitting AIDS or viral hepatitis; known hypersensitivity or contraindications to ACE inhibitors, nifedipine, or atenolol; women of childbearing age not using medically acceptable methods of birth control (oral contraceptives were not allowed) or those planning a pregnancy during the treatment period; treatment compliance over the 4-week placebo run-in of <85%; patient on antihypertensive treatment.

Randomization (visit 2)

A total of 137 patients met these clinical criteria, were assigned the next available trial number, and were randomized, in a double-blind fashion, to one of three treatment groups: 47 to lisinopril, 41 to nifedipine, and 49 to placebo. A double-dummy technique was used throughout the study to maintain blindness. Treatment was allocated in balanced blocks. Of the 137 patients, 14 on lisinopril, 13 on nifedipine, and 9 on placebo were excluded for not having AER values within 20–200 μg/min (according to the central laboratory measurement). Two patients on lisinopril, one on nifedipine, and six on placebo were excluded: one became pregnant, three requested exclusion, one had burns, two had lower-limb edema, one had hyperkalemia, and one (on nifedipine) experienced sudden death. Lisinopril (Zestril, Zeneca, Milan, Italy) was supplied as 2.5-, 10-, and 20-mg tablets with matching placebo. Nifedipine (Adalat Retard, Bayer, Milan, Italy) was supplied as 10- and 20-mg capsules with matching placebo. Both active medications and placebo were supplied by Zeneca Pharmaceuticals.

All patients received one lisinopril placebo tablet once daily and one nifedipine placebo capsule twice daily for 4 weeks. On the day of randomization, patients received one 2.5-mg tablet of lisinopril (if randomized to the lisinopril treatment group) or matching placebo as a single initial dose and remained under the investigator's observation for 4–5 h after the dose. Patients randomized to the nifedipine group received the initial dose of nifedipine (10 mg) in the evening. On the day after the single initial dose, patients started low-dose active treatment with lisinopril (10 mg), nifedipine (10 mg), or placebo. Treatment was taken twice daily, with patients receiving one tablet and one capsule in the morning and one capsule in the evening.

Inclusion visit (visit 3) and follow-up procedures

A total of 92 patients were included in the study according to the parameters explained above. If at any visit, starting 1 month after randomization (visit 3), BP was not adequately controlled (a reduction in both standing sBP and standing dBP of <5% of baseline values), the low dose of lisinopril, nifedipine, or placebo had to be doubled (step 2). Patients were instructed to take their morning dose on the day of a clinic visit when the visit was over. If at any visit, starting 3 months after randomization (from visit 5), BP was not adequately controlled despite treatment with the higher dose (i.e., a reduction in both standing sBP and standing dBP of <5% of baseline values and standing BP >140/90 mmHg at two consecutive visits on the higher dose), 50 mg o.d. of atenolol was added to the ongoing treatment (step 3). If BP was not adequately controlled (standing BP >140/90 mmHg) at any visit after the introduction of 50 mg o.d. of atenolol, the atenolol dose was increased to 100 mg o.d. If BP was not adequately controlled (standing BP >160/90 mmHg) at any subsequent visit, the patient had to be withdrawn.

The duration of treatment was to be 3 years. At each visit, the patient was supplied with two pots containing the treatment doses for the subsequent period (with 25% excess). For each patient, a sealed randomization code was supplied to the clinician. Sealed randomization codes were returned to Zeneca Pharmaceuticals at the end of the study. Patients were asked to return all the remaining tablets in order for us to check their compliance with therapy. The following drugs were not allowed: drugs affecting lipid levels, aldose reductase inhibitors, steroids or oral contraceptives, gold, penicillamine, nonsteroidal anti-inflammatory agents, MAO (monoamine oxidase) inhibitors, lithium, tricyclic antidepressants, and cimetidine. The occasional administration of paracetamol as a mild analgesic was allowed.

Clinical methods

Patients continued to take their usual insulin for glycemic control. No patient needed more than 80 U of insulin per day. Arterial BP (phase I/V) was measured by mercury sphygmomanometer with appropriately sized cuff at least 2 h after the administration of insulin and study medication in the morning after breakfast. Three readings were taken 5 min apart to the nearest 2 mmHg after the patient had been sitting quietly for at least 15 min, and three more readings were taken after 15 min of standing. The average of the three measurements in each position was used in the analysis. Clinical

Table 1—*Demographic and clinical features of 92 IDDM patients with incipient nephropathy*

	Placebo group	Lisinopril group	Nifedipine group
n (M/W)	23/11	21/11	19/7
Age (years)	37 ± 10	38 ± 11	36 ± 10
Weight (kg)	67 ± 9	68 ± 9	68 ± 10
Height (cm)	168 ± 7	167 ± 9	168 ± 8
Duration of IDDM (years)	19 ± 8	19 ± 8	17 ± 6
Smoker (yes/no)	20/14	18/14	15/11
ECG abnormalities (yes/no)	7/34	2/32	2/26

Data are means ± SD for patients who participated in the final examination after visit 3. Patients who were protocol violators or who dropped out for other reasons, despite randomization, were not included. Electrocardiogram (ECG) abnormalities include features of stable ischemia, previous myocardial infarction, atrial enlargement, and/or left ventricular hypertrophy; patients with chronic arrythmias were excluded.

evaluations were performed every 6 months for 3 years. Three timed overnight urine samples were collected for AER determination at each follow-up visit. Patients followed an isocaloric diet (1,500 kcal/1.73 m^2 daily) containing no simple carbohydrates, 1.0 g/kg protein daily, <6 g NaCl/day, and <30% fat content with a higher percentage of unsaturated fatty acids. Smoking was not a criterion for exclusion.

Laboratory methods

Urinary albumin concentration was measured by radioimmunoassay (RIA) in one reference laboratory, where urine specimens were routinely sent and stored at −20°C (10,11). Usually, urine specimens were thawed only once before measurement. The inter- and intra-assay coefficients of variation for urinary albumin concentration were 11 and 4%, respectively (11). GFR was determined by the modeling analysis of the plasma decay of ^{51}Cr-labeled EDTA; frequent blood samples were obtained within 300 min after abrupt intravenous injection of the tracer compound (12). Concentrations of serum and urine creatinine were measured according to a modification of the method of Jaffé using a Technicon SMA II C Autoanalyzer (Technicon, Tarrytown, NY) (13). Creatinine clearance was calculated from serum values according to the formula of Cockcroft and Gault (14). HbA_{1c} was measured by high-pressure liquid chromatography (15). Triglycerides, cholesterol, HDL cholesterol (16), sodium, and potassium concentrations were measured in each center as described elsewhere (13).

Outcome variables

The primary outcome variable was the progression rate of MA to persistent macroalbuminuria defined as an AER value >200 µg/min in two out of three or more consecutive visits. The secondary outcome variables were *1*) the number of patients who had a 50% yearly increase in AER above baseline values and *2*) the number of patients who showed a regression rate from MA to persistent normoalbuminuria defined as an AER value <20 µg/min in two out of two or more consecutive visits.

Statistical analysis

Continuous and dichotomous baseline demographic, clinical, and laboratory characteristics of the groups with active treatments (lisinopril and nifedipine) and placebo were analyzed using Fisher's exact tests and by analysis of variance (17). Progression from MA to macroalbuminuria, regression from MA to normoalbuminuria, and rate of progression of AER >50% per year above baseline values were analyzed as dichotomous variables using the Cox proportional hazards model, which estimates the risk of reduction associated with an intervention trial adjusted for confounding variables: number of patients examined each year, duration of follow-up when the clinical event developed, HbA_{1c} levels, BP levels, and baseline AER (18). In particular, the average relative risk (RR), comparing either the lisinopril or the nifedipine treatment group with placebo over the entire period of observation, was estimated by a proportional hazards regression model adjusted for numerical baseline values. The Wald test from the proportional hazards model was used to test the difference between cumulative incidence curves. The adjusted percentage reduction in risk for active therapy versus placebo was calculated from the average adjusted RR of active therapy in comparison with placebo as (1-RR) × 100%. The length of time-to-event endpoints, product-limit life-table distributions, were compared with the log-rank statistics. Wilcoxon's rank-sum test was used to compare the treatment groups with respect to the distributions of numerical variables, and the contingency χ^2 test was used for categorical variables (19). For the analysis of continuous variables at each follow-up visit, the change of variables from baseline was assessed and the multivariate Wilcoxon analysis was used in conjunction with an overall test of differences between groups using weights proportional to the total sample size at each annual evaluation (20). Statistical analysis was carried out using SPSS 6.1 Base System software (SPSS, Chicago, IL).

RESULTS — During the 3-year, double-blind, double-dummy, randomized study, 10 of the 92 patients discontinued the study. One patient who was on lisinopril treatment requested to be withdrawn, another patient on lisinopril was withdrawn for HbA_{1c} values repeatedly >12%, two patients who were on nifedipine treatment were withdrawn at their request, and six patients who were on placebo were withdrawn for herpes zoster, lung cancer, flatulence, tuberculosis, severe diabetic neuropathy, and myocardial infarction, respectively. Twenty percent of those on placebo, 21% of those on lisinopril, and 27% of those on nifedipine also had concomitant untoward side effects that did not cause withdrawal from the study. It has to be pointed out that postural arterial hypotension was neither observed in any patient nor the cause of withdrawal from the study of any patient.

The baseline demographic, clinical, and laboratory data of the groups randomized to placebo, lisinopril, and nifedipine are shown in Tables 1, 2, and 3. Patients were reasonably well-matched, particularly with regard to AER, serum creatinine, HbA_{1c}, and serum lipids. GFR values were also similar in the three groups of patients.

During the 3-year follow-up period, 7 of 34 placebo-treated (20.6%), 2 of 32 lisinopril-treated (6.3%), and 2 of 26 nifedipine-treated (7.7%) patients progressed to clinical albuminuria (Fisher's exact test, $P < 0.03$). Time-to-event analysis indicated a reduction in the risk of progression to macroalbuminuria of 49.1% (95% CI 26.8–63.4%) in the lisinopril ($P < 0.03$) and 41.4% (95% CI 22.1–64.3%) in the nifedipine ($P < 0.05$) groups, respectively, after adjustment for sBP, dBP, HbA_{1c}, and baseline AER (Fig. 1). Baseline AER

Table 2—*AER, GFR, serum creatinine, creatinine clearance (calculated from serum creatinine), serum potassium, and albumin in 92 IDDM patients with incipient nephropathy*

	Placebo group		Lisinopril group		Nifedepine group	
	Baseline	3rd year	Baseline	3rd year	Baseline	3rd year
AER (μg/min)	88 (21–187)	119 (4–176)	54 (20–128)	29 (5–173)†	70 (28–174)	58 (2–198)*
GFR (ml · min^{-1} · 1.73 m^{-2})	107 ± 20	—	122 ± 14	—	118 ± 16	—
Serum creatinine (μmol/l)	88 ± 14	95 ± 19	86 ± 13	77 ± 16	83 ± 11	86 ± 10
Creatinine clearance (ml · min^{-1} · 1.73 m^{-2})	110 ± 15	105 ± 15	113 ± 16	109 ± 19	105 ± 14	101 ± 15
Serum potassium (mmol/l)	4.1 ± 0.4	4.4 ± 0.4	4.2 ± 0.3	4.5 ± 0.4	4.2 ± 0.5	4.4 ± 0.3
Albumin (g/dl)	4.4 ± 0.2	4.4 ± 0.3	4.4 ± 0.4	4.3 ± 0.3	4.4 ± 0.3	4.2 ± 0.4

Data are means ± SD or median (range). Baseline values are for 92 patients who participated to the final statistical analysis after visit 3; 3rd-year values are for the 82 patients who completed the study. *$P < 0.03$; †$P < 0.003$.

was 71.1 μg/min (range 20.7–187.3) in progressors and 73.3 μg/min (range 20.2–174.1) in nonprogressors (NS).

We also calculated the regression rate from MA to normoalbuminuria. During the 3-year follow-up period, 4% of the placebo-treated, 15% of the lisinopril-treated, and 0% of the nifedipine-treated patients showed a regression to normoalbuminuria (Fisher's exact test, $P < 0.001$, lisinopril vs. placebo; NS, nifedipine vs. placebo). The percentage of patients who showed a >50% yearly increase of AER above baseline values was significantly lower in the lisinopril group (13 of 32, 40.6%), but not in the nifedipine group (15 of 26, 57.7%), than in the placebo group (23 of 34, 67.6%) (Fig. 2) (nifedipine vs. placebo: $P < 0.39$; lisinopril vs. placebo: $P < 0.02$; lisinopril vs. nifedipine: $P < 0.17$).

Creatinine clearance, calculated from serum creatinine concentrations, was similar in the three groups at baseline and after the 3-year follow-up period. Table 3 shows the mean ± SD sBP and dBP levels in the placebo, lisinopril, and nifedipine groups measured in a sitting position at baseline and at the first, second, and third years of follow-up. Baseline sBP and dBP were similar in the three groups. dBP levels were slightly (~3–6 mmHg), albeit not significantly, lower during the follow-up period in the lisinopril group than in either the nifedipine or the placebo group. The lisinopril group had significantly lower sBP values during follow-up than either the nifedipine ($P < 0.05$) or the placebo ($P < 0.01$) group. Both sBP and dBP levels were related to progression rate of MA to macroalbuminuria, to regression of MA to normoalbuminuria, and to the absolute change of AER during the follow-up period. Of patients on placebo, on lisinopril, and on nifedipine, 27, 58, and 33%, respectively, required the low-dose treatment; 68, 37, and 61%, respectively, required the high-dose treatment; and 5, 5, and 6%, respectively, required atenolol.

HbA_{1c} levels were similar in the three groups at baseline (nifedipine vs. lisinopril vs. placebo: 7.9 ± 1.6 vs. 8.2 ± 1.5 vs. 8.9 ± 1.6%; mean ± SD). No change was found during the follow-up period. No differences were observed among the three groups during the 3-year follow-up with regard to circulating levels of cholesterol, triglycerides, LDL cholesterol, HDL cholesterol, and lipoprotein. No significant change was found in the three groups with regard to serum albumin and potassium concentrations during the follow-up period. In particular, no patient withdrew because of hyperkalemia.

CONCLUSIONS — The rate of occurrence of macroalbuminuria (>200 μg/min) may be used as an outcome measure in intervention trials in IDDM patients with incipient nephropathy (21,22). In fact, the occurrence of overt proteinuria strongly predicts the development of end-stage renal disease in IDDM (6,7). In addition, a yearly increase of AER above baseline values could be helpful in evaluating the effects of antihypertensive therapy on the progression of kidney damage in diabetic patients with incipient nephropathy with or without arterial hypertension (8). The rationale for choosing this latter outcome measurement is supported by the observation during intervention trials that patients in whom AER at baseline is lower have a better prognosis for the progression rate of AER and renal function (23). In particular, it has been postulated that albumin and protein leakage through the glomerular barrier itself is one of the factors that contribute to the progression of renal damage (23).

We confirm the results of recent trials demonstrating that ACE inhibitors lead to a marked reduction in the risk of progression to macroalbuminuria in IDDM patients with MA (24,25). The novel finding in our study is that both the long-acting ACE inhibitor lisinopril and the slow-release calcium channel blocker nifedipine are capable of significantly delaying the progression of MA toward macroalbuminuria in IDDM patients with incipient nephropathy and normal BP. However, lisinopril appears

Table 3—*Standing sBP and dBP in IDDM patients with incipient nephropathy*

	Placebo group			Lisinopril group			Nifedipine group		
	Baseline	2nd year	3rd year	Baseline	2nd year	3rd year	Baseline	2nd year	3rd year
sBP	129 ± 8	121 ± 10	126 ± 12	126 ± 8	118 ± 11	117 ± 11	131 ± 10	129 ± 13	129 ± 15
dBP	83 ± 4	77 ± 7	76 ± 7	82 ± 5	76 ± 7	73 ± 7	84 ± 4	79 ± 7	79 ± 7

Data are means ± SD. Baseline and 2nd-year values are for 92 patients; 3rd-year values are for the 82 patients who completed the 3rd year. For the lisinopril group, $P < 0.01$ lisinopril vs. placebo; $P < 0.05$ vs. nifedipine.

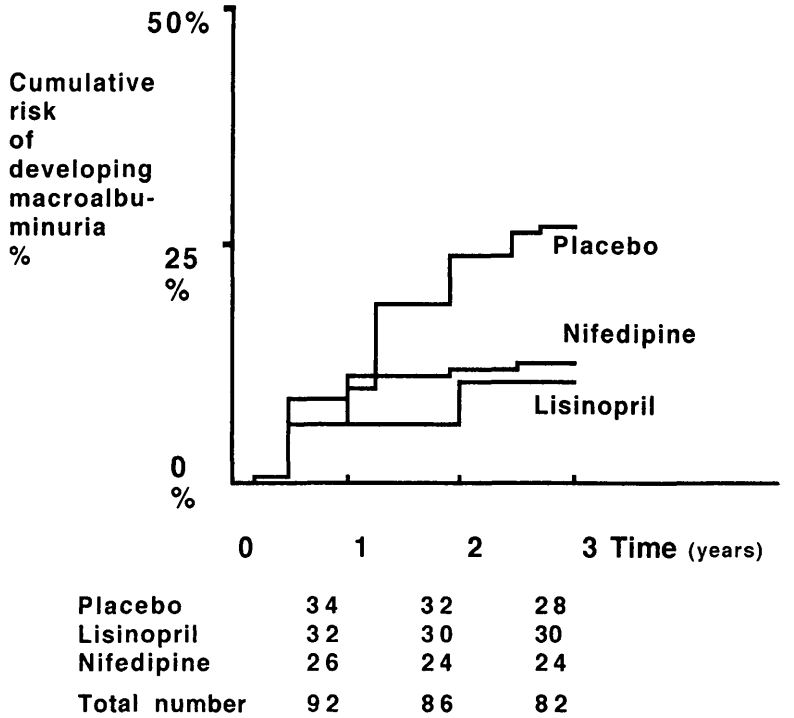

Figure 1—*Cumulative risk of progression to macroalbuminuria from MA in normotensive IDDM patients with incipient diabetic nephropathy during placebo, lisinopril, and nifedipine treatments. The numbers at the bottom of the figure indicate the patients in each group at risk for progression to macroalbuminuria at baseline and after 1 and 2 years of follow-up.*

more effective than nifedipine in slowing the course of nephropathy.

Long-term studies have examined the effects of ACE inhibitors and calcium channel blockers on kidney function in hypertensive NIDDM patients with incipient nephropathy (26–29). These reports showed that calcium channel blockers prevented the GFR fall in a fashion similar to that of ACE inhibitors, although the decrease of AER was significantly greater with ACE inhibitors.

The mechanisms by which calcium channel blockers and ACE inhibitors modulate renal and vascular function are different (24,25). Studies in animals have suggested that ACE inhibitors, but not calcium channel blockers, decrease intraglomerular pressure (24,25). ACE inhibitors blunt or suppress the vasoconstrictor, hypertrophic, and hyperplastic effects of angiotensin II on mesangial and vascular smooth muscle cells. Studies in humans have shown no consistent differences in the degree of such inhibition caused by different ACE inhibitors (24,25).

The favorable effect of calcium channel blockers may be related to the modulation of intracellular free calcium concentrations, which are altered in patients with essential hypertension and with diabetes (30–32). Furthermore, calcium channel blockers inhibit the effects of angiotensin II, but also of other mitogens, on renal function (33). Other researchers have suggested that the effects of calcium channel blockers on AER are heterogeneous and, in particular, that nonhydropyridine, but not dihydropyridine, calcium channel blockers lower AER (33,34). The reason for such differences would be that nonhydropyridine drugs block different calcium channels than dihydropyridine drugs, resulting in a different modulation of hemodynamic function and growth of mesangial and vascular smooth muscle cells (33).

The results of the present study, as well as those of a previous report by our research group comparing ACE inhibitors and amlodipine in NIDDM (28), suggest that dihydropyridine calcium channel blockers are also capable of ameliorating kidney function to an extent similar to that of ACE inhibitors. However, our data show that lisinopril was more effective than nifedipine with regard to the regression rate of MA to normoalbuminuria and to the number of patients with a yearly increase of AER >50% above baseline values. These latter findings deserve emphasis, because it is likely that abnormalities in AER not only reflect the occurrence of renal disorders but also contribute to the cause and progression of renal damage (23).

Our studies are in keeping with the conclusions of two recent meta-analysis studies showing a more significant reduction of AER during therapy with ACE inhibitors than with other therapeutic agents. This markedly noticeable effect has been attributed to a peculiar capacity of ACE inhibitors to decrease glomerular permeability to albumin. BP levels were significantly lower in the lisinopril group, although the daily dose of ACE inhibitors used during the double-blind, double-

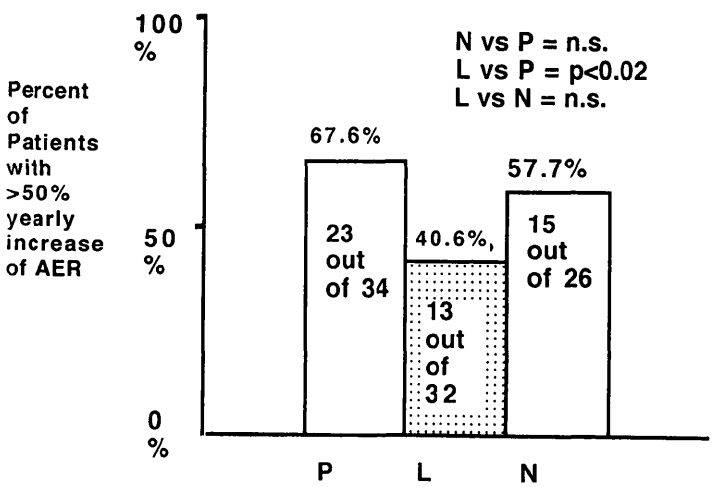

Figure 2—*Percentage and absolute numbers of IDDM patients with incipient nephropathy who showed a >50% yearly increase of AER from baseline during the 3 years of follow-up. P, placebo; L: lisinopril; and N, nifedipine.*

dummy, tapering stepwise procedure was significantly reduced in comparison with that of the nifedipine group. Slightly lower levels of BP during treatment with ACE inhibitors are often observed in clinical trials comparing ACE inhibitors with other therapeutic compounds, despite blind treatment and similar efforts dedicated to achieve BP control (24,25). This small, but almost constant, difference in BP levels hinders the interpretation of the precise mechanisms accounting for the amelioration of renal function during treatment with ACE inhibitors in comparison with other therapeutic approaches. In fact, this improvement could be due to either the lower BP levels or a peculiar action of ACE inhibitors on the kidney, irrespective of their hemodynamic effect.

In an attempt to circumvent the confusing interaction of these two variables, we evaluated the risk of progression toward macroalbuminuria after adjustment for BP patterns, because they were significantly lower during lisinopril treatment than both nifedipine and placebo treatments. This risk of progression toward macroalbuminuria was, however, significantly lower during both nifedipine and lisinopril treatments than during placebo treatment, even taking into account the difference in BP levels.

In conclusion, our data show that both lisinopril and nifedipine are effective in delaying the occurrence of macroalbuminuria in normotensive IDDM patients with MA. Because overt proteinuria strongly predicts end-stage renal disease, both treatments appear capable of delaying the occurrence of such complications in normotensive IDDM patients with incipient nephropathy. However, lisinopril appears more powerful than nifedipine in slowing the course of nephropathy. This favorable outcome is not accompanied by untoward side effects, such as postural hypotension and hyperkalemia. Lisinopril improves the patterns of AER in the range of MA and increases the percentage of patients who become normoalbuminuric. This latter result needs to be emphasized, because the patterns of AER have been suggested to be not only an indicator of but also a contributor to the progression of renal damage.

APPENDIX

The Italian Microalbuminuria Study Group (IMSG)

Coordinator: G. Crepaldi; participating diabetic outpatient clinics and centers: Endocrinology and Department of Clinical Medicine, Policlinico Universitario Umberto I, Universita' La Sapienza, Rome: D. Andreani, P. Pietravalle, S. Morano; University of Palermo: G.D. Bompiani; Dipartimento di Medicina Interna e Scienze Endocrine e Metaboliche, Universita' di Perugia: P. Brunetti, F. Santeusanio, A. Bueti; Diabetolgia, Azienda Ospedaliera San Giovanni Battista di Torino: Q. Carta, G. Petraroli, U. Giovannelli; National Research Center for the Study of Ageing, Istituto di Medicina Interna, Universita' di Padova e Diabetic Outpatient Clinics of Monfalcone, Mestre, Contarina ed Este: M.R. Cipollina, A. Carraro, P. Fioretto, I. Barzon, E. Brocco, C. Abaterusso, L. Occhionero, M. Velussi, F. Frigato, F. Piarulli, M. Sambataro, B. Muollo, A. Saller; Dipartimento di Medicina Interna, DIMI dell' Universita' di Genova: G. Deferrari, S. Saffioti, L. Turrini Dertenois; Department of Clinical Medicine, Universita' di Brescia: G. Romanelli, G. Giustina, A. Giustina; Endocrinology and Metabolic Diseases, Universita' di Parma: A. Gnudi, R. Lugari; Endocrinology and Metabolic Diseases, Universita' di Pisa: R. Navalesi, G. Penno, L. Rizzo; Division of Endocrinology, Ospedali Riuniti di Bergamo: G. Pagani, L. Cortesi, A. Pedroncelli; Istituto Scientifico S. Raffaele, Universita' di Milano: R. Ghelardi, E. Pezzali, R. Mangili, G. Pozza; Central Laboratory for Urinary Albumin Excretion Measurements, Istituto Scientifico S. Raffaele, Universita' di Milano: G. Meregalli, D. Gabellini, F. Garbetta; Istituto di Ricerche Farmacologiche Mario Negri, Laboratorio Mario Negri Bergamo: G. Remuzzi, P. Ruggenenti, E. Daina; Endocrinology and Metabolic Diseases, Universita' di Bari: R. Giorgino, A. Bellomo Damato, E. Turrisi; Diabetologia e Malattie del Ricambio, Ospedale Civile San Salvatore, Pesaro: A. Spalluto, I. Trau, L. Maggiulli; Dipartimento di Medicina Interna, Universita' di Napoli: M. Mancini, I. Gaeta, L.A. Ferrara, G. Riccardi; Centro per le Malattie Dismetaboliche, Ospedale Civile di Cagliari: S. Muntoni, M. Songini; Endocrinologia e Malattie del Ricambio Universita' di Firenze: G.M. Rotella, F. Piani.

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